Date: 15/10

CHAPTER

SIGNAL GENERATOR

SINE WAVE GENERATOR (OSCILLATOR)

OSCILLATORS:

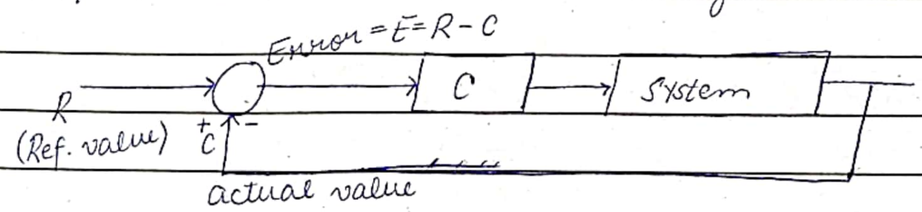
Everything having oscillator - bi-directional signal

Oscillators generates a finite output signal by using an amplifier followed by a "frequency selective circuit" connected in +ve feedback loop with NO input signal

freq. selective circuit
⇒ filter

NEGATIVE FEEDBACK LOOP

(it is preffered because error converges to zero - stable system)



REF TEMP. (R)	ACTUAL TEMP. (C)	ERROR (E)
25°	30°	−5°
25°	27°	−2°
25°	22°	+3
25°	26°	−1
25°	24.5°	+0.5
25°	25.2°	−0.2
25°	25°	0
25°	25°	0

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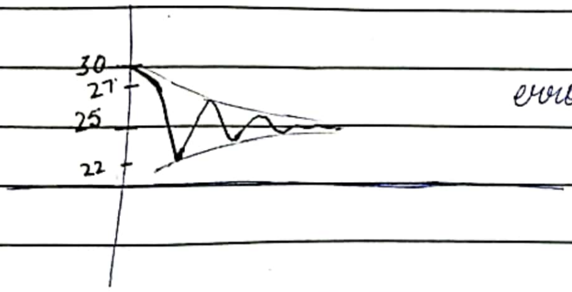
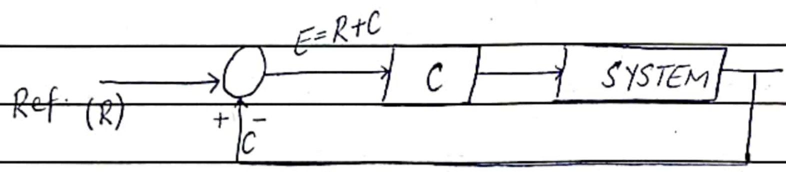
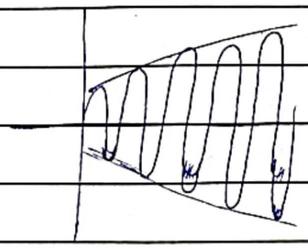


error converging to zero

POSITIVE FEEDBACK SYSTEM

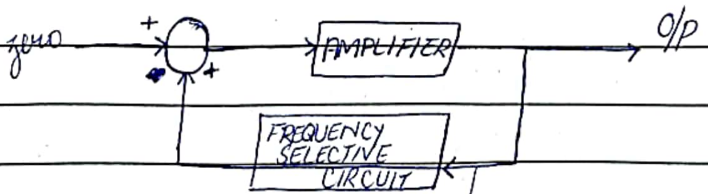


REF. (R)	ACTUAL TEMP (C)	ERROR (E=R+C)
25°	30°	+55°
25°	35°	+60
		⋮
		∞



(exponentially increasing)
error diverging – unstable system

SINUSOIDAL OSCILLATOR:



Date:

SUPPOSE:

$v_i = x_s$

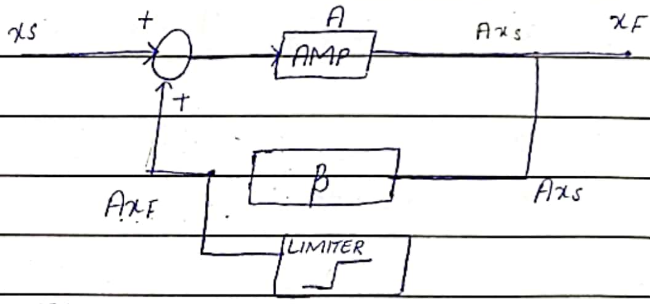
x_s — reservoir of infinite sinusoidal frequency.

$x_s = M_1 \sin(\omega t) + M_2 \sin(2\omega t) + M_3 \sin(3\omega t) + \dots + M_f \sin(f\omega t) + \dots + M_n \sin(n\omega t)$ ($M_f \sin(f\omega t) \rightarrow x_F$)

* magnitude of $M_1, M_2, M_3, \dots, M_n$ — is extremely small of the order 10^{-30}

amplifier's gain — A, freq. selective circuit — β

O/P = finite signal $= x_F$



(Sum) → (amp) amp's gain → (Freq.)

I — $x_s + 0 \longrightarrow Ax_s \longrightarrow Ax_F\beta$

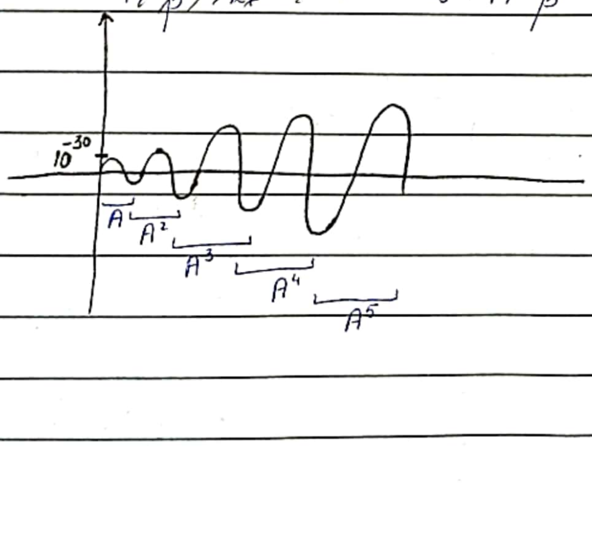
II — $x_s + A\beta x_F \longrightarrow Ax_s + A^2\beta x_F \longrightarrow A^2\beta^2 x_F$

III — $x_s + A^2\beta^2 x_F \longrightarrow Ax_s + A^3\beta^2 x_F \longrightarrow A^3\beta^3 x_F$ $\quad \therefore A = 10^5, \beta = 1$

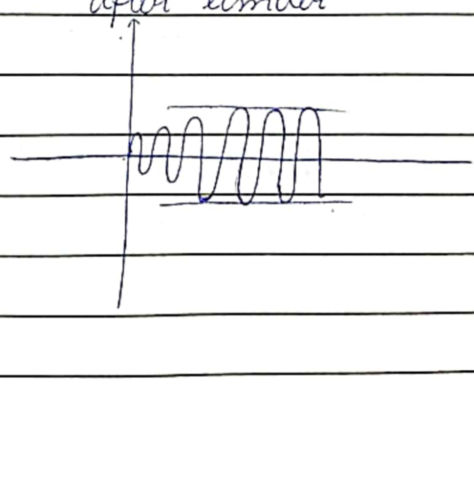
IV — $x_s + A^3\beta^3 x_F \longrightarrow Ax_s + A^4\beta^3 x_F \longrightarrow A^4\beta^4 x_F$

$\downarrow \qquad \downarrow \qquad \downarrow$

$A^n\beta^{n-1} x_F \longrightarrow Ax_s + A^n\beta^{n-1} x_F \longrightarrow A^n\beta^n x_F$



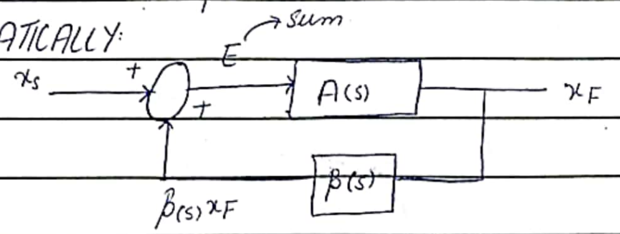
after limiter



Date:

Thermal noise of the resistor is used

MATHEMATICALLY:



$$E = x_s + \beta(s)\, x_F \qquad (i)$$

$$x_F = A(s)\, E \qquad (ii)$$

Put (i) in (ii)

$$x_F = A(s)\left(x_s + \beta(s)\, x_F\right)$$

$$x_F \left(1 - A(s)\beta(s)\right) = A(s) \times x_s$$

$$\boxed{\frac{x_F}{x_s} = \frac{A(s)}{1 - A(s)\beta(s)}}$$

When NO I/p is connected

$$\frac{x_F}{x_s} = \frac{x_F}{0} = \infty$$

$$\frac{A(s)}{1 - A(s)\beta(s)} = \infty \qquad \text{(when denominator = 0)}$$

$$\Rightarrow 1 - A(s)\beta(s) = 0$$

$$A(s)\beta(s) = 1 \qquad \text{(BORK HAUSAN'S CRITERIA)}$$

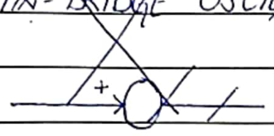
$$\therefore A(\omega_0)\beta(\omega_0) = 1$$

$$\omega_0 = 2\pi f$$

Date:

So, if we apply zero input, the output would be a finite signal of frequency (f_0) if and only if $A(\omega_0)\beta(\omega_0) = 1$; $\omega_0 = 2\pi f$ holds at the frequency (f_0)

WEIN-BRIDGE OSCILLATOR:



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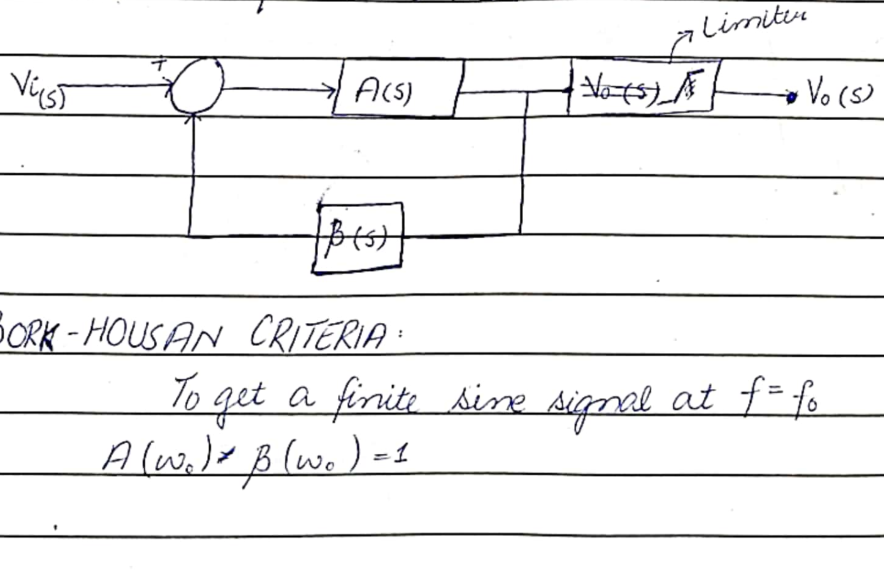
WEIN-BRIDGE OSCILLATOR



BORK-HOUSAN CRITERIA:

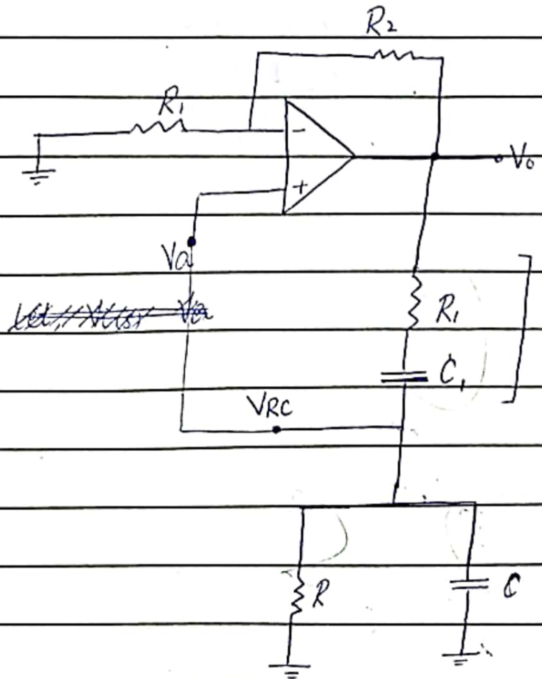
To get a finite sine signal at $f = f_0$

$A(\omega_0) \times \beta(\omega_0) = 1$

Date:

circuit when limiter is not connected (as previous)

(-ve feedback (O/p connected to -ve terminal))

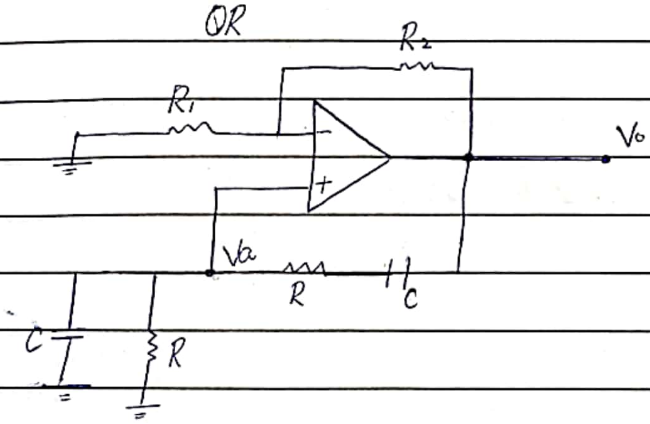


Let, $V_{i(s)} = V_a$

$\Rightarrow$ ~~$\boxed{A(s)}$~~

Remove 'V_a' & 'V_{RC}' after joining names

OR



(non-inv.)

$V_{o(s)} = \left(1 + \frac{R_2}{R_1}\right) V_{a(s)}$

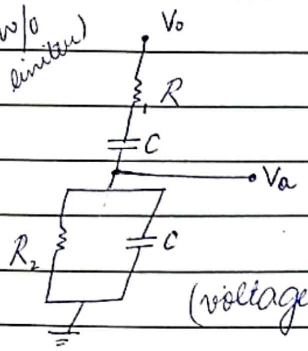
$\frac{V_{o(s)}}{V_{a(s)}} = \left(\frac{1 + R_2}{R_1}\right)$

$\frac{V_{o(s)}}{V_{a(s)}} = \cancel{A} A(s)$

$\therefore A(s) = 1 + \frac{R_2}{R_1}$

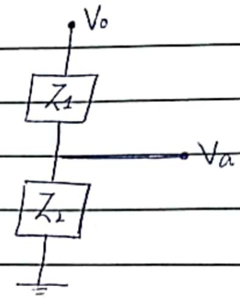
Date:

(w/o limiter)

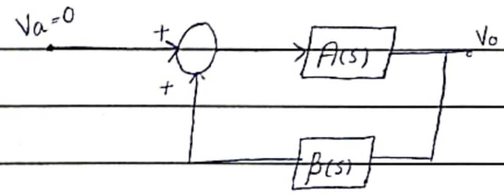


(voltage divider)

$\Longrightarrow$



$$\beta(s) = \frac{Z_2}{Z_1 + Z_2}$$



$$= \frac{R_2 \parallel (1/sC)}{\left(R_1 + \frac{1}{sC}\right) + \left(R_2 \parallel \frac{1}{sC}\right)}$$

$$= \frac{1 + \cancel{(R_2/R_1)}}{3 + (RC)s + \frac{1}{(RC)s}}$$

$$\therefore \quad A(s) = 1 + \frac{R_2}{R_1}$$

* $A(s)\,\beta(s) = 1$,

$$A(s)\,\beta(s) = \frac{1 + (R_2/R_1)}{3 + (RC)s + 1/(RC)s}$$

Put $s = j\omega$

$$A(\omega)\,\beta(\omega) = \frac{1 + (R_2/R_1)}{3 + j(RC\omega) + \frac{1}{j(RC\omega)}}$$

Date:

$$= \frac{1 + (R_2/R_1)}{3 + j\left(RC\omega - \frac{1}{RC\omega}\right)}$$

BARK HOUSAN CRITERIA: To get a 'REAL finite' sine wave of freq. f_0,

$A(\omega)B(\omega) = 1$ at $f = f_0$

$$A(\omega_0)\beta(\omega_0) = \frac{1 + (R_2/R_1)}{3 + j\left(RC\omega_0 - \frac{1}{RC\omega_0}\right)}$$

If we make this imaginary part $= 0$, then this whole thing becomes REAL.

TO MAKE SYSTEM REAL:

$$\frac{RC\omega_0^2 - 1}{RC\omega_0} = 0$$

$$RC\omega_0 = \frac{1}{RC\omega_0} \qquad \therefore \qquad 2\pi f_0 = \omega_0$$

$$\omega_0^2 = \frac{1}{(RC)^2} \Rightarrow \boxed{\omega_0 = \frac{1}{RC}} \quad \text{(filter design)}$$

$\because$ FOR R_1 & R_2

$$A(\omega_0)\beta(\omega_0) = 1$$

$$\frac{1 + (R_2/R_1)}{3 + j\left(RC\omega_0 - \frac{1}{RC\omega_0}\right)} = 1 \qquad \therefore \text{ at } \omega_0 = \frac{1}{RC}$$

$$\frac{1 + (R_2/R_1)}{3} = 1$$

EAGLE

Date:

$1 + \frac{R_2}{R_1} = 3$

$\frac{R_2}{R_1} = 2$; * $\boxed{\frac{R_2}{R_1} \geq 2}$ Assume, $R_1 = 100k$

$R_2 = 200k$ ↓ should be greater (practically)

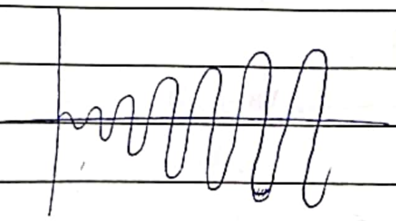
$\Rightarrow R_2 = 201 k\Omega$

★ FILTER:

$\omega_0 = 1/RC$, $\omega_0 = 2\pi f_0$; $f_0 \rightarrow$ given

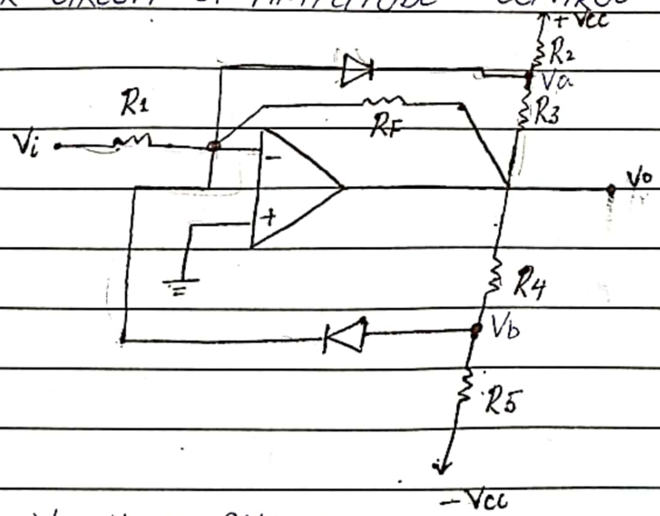
AMP:

$R_2/R_1 \geq 2$



(W/O limiter)

LIMITER CIRCUIT OF AMPLITUDE CONTROL:



FOR +VE HALF CYCLE

• V_i is very small (close to zero) but +ve.

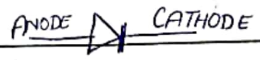
• $V_o = -\left(\frac{R_F}{R_1}\right) \times V_i$ (inverting amplifier)

Date: ______

- V_o is also very small and -ve ≈ -0.01

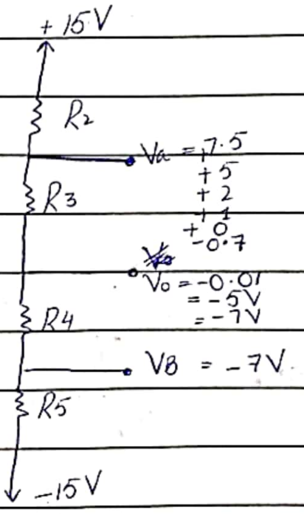
*[$V_{ANODE} > V_{CATHODE}$ (short $\overset{\nearrow \text{forward}}{\text{circuit}}$)

[$V_{ANODE} < V_{CATHODE}$ (open circuit) — reverse biased)

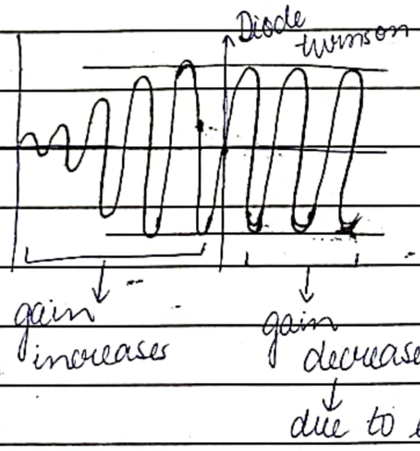


- V_i will gradually increase
- V_o will more & more -ve ~~(decrease)~~ (↑)

$$\Rightarrow V_o = -\left(\frac{R_F}{R_1}\right) \times V_i$$



- V_A will continue to decrease in magnitude
- Until $V_A = -0.7V \rightarrow$ at this point 'D_1 turns on'
- It brings R_3 in parallel with R_F.
  - $V_o = -\left(\frac{R_3 \parallel R_F}{R_1}\right) \times V_i$



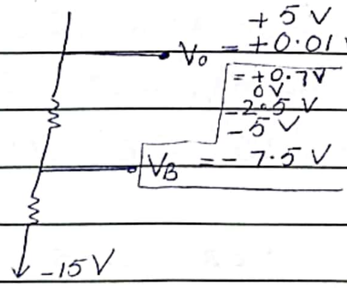
$$* \frac{R_F}{R_1} >> \frac{R_3 \parallel R_F}{R_1}$$

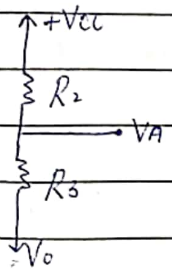
$$G_1 >> G_2$$

Date:

FOR -VE CYCLE:

- V_i becomes more '-ve'
- V_o becomes more '+ve'
- V_B ~~is~~ becomes less '-ve'
- $V_o = -\left(\frac{R_F \parallel R_4}{R_1}\right) \times V_i$ (diode short)

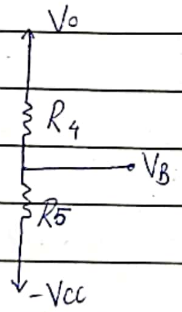




$$V_A = \left(\frac{R_3}{R_2+R_3}\right)V_{cc} + \left(\frac{R_2}{R_2+R_3}\right)V_o \qquad (i)$$

$\underbrace{\left(\frac{R_3}{R_2+R_3}\right)V_{cc}}_{V_o\ \text{~~ground~~}}$ $\underbrace{\left(\frac{R_2}{R_2+R_3}\right)V_o}_{V_{cc}\ —\ \perp}$

~~V_A~~



$$V_B = \left(\frac{R_4}{R_4+R_5}\right)(-V_{cc}) + \left(\frac{R_5}{R_4+R_5}\right) \times V_o \qquad (ii)$$

for (i)

~~V_A~~ $V_A = -V_D$, $V_o = L_-$

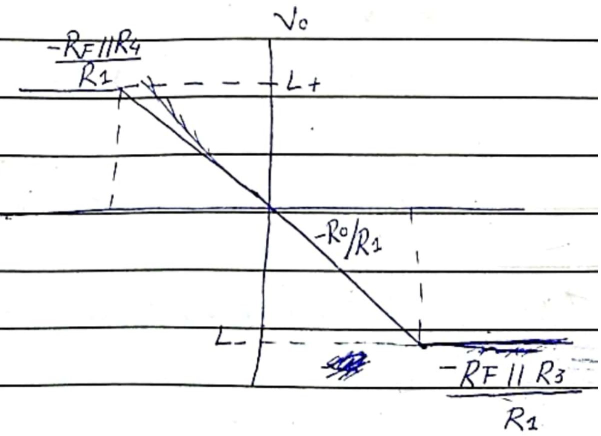
$\Rightarrow$ (i) $L_- = -V_{cc} \times \frac{R_3}{R_2} - V_D\left(1 + \frac{R_3}{R_2}\right)$

for (ii)

$V_B = +V_D$, $V_o = L_+$

$\Rightarrow$ (ii) $L_+ = +V_{cc}\left(\frac{R_4}{R_5}\right) + V_D\left(1 + \frac{R_4}{R_5}\right)$

TRANSFER CHARACTERISTICS



Date: 22/10

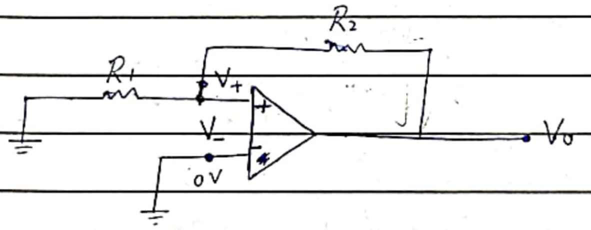
MULTI-VIBRATOR CCt's FOR SIGNAL GENERATION

used to generate sq. wave

- Bistable MVB (2 stable states)
- ~~Mono~~-stable MVB (1 stable states) (used in alarms)
- A-Stable MVB (No stable state)

→ BISTABLE MVB:

- Bistability: DC amplier in +ve feedback loop (no filter)
- *Open loop gain: $A\beta > 1$
- Feed back contain a voltage divider of R_1 & R_2



- $V_- = 0V$
- No extited input voltage
- Electrical noise in cct introduces a small +ve voltage at V_+

Hence,

$$V_o^1 = A(\overset{+ve}{V_+} - \overset{0V}{V_-})$$

$$\Rightarrow V_o^1 > 0$$

$V_o^{(1)} \rightarrow 1^{st}$ iteration

O/P is fed back at $V_+ = \underbrace{\left(\frac{R_1}{R_1 + R_2}\right)}_{\beta} \times V_o^1$

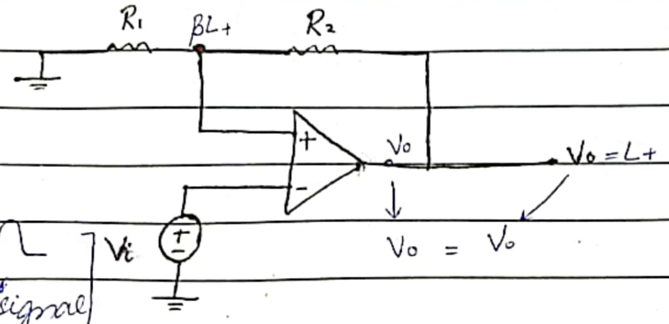
$$V_+ = \beta V_o^1$$

*iterations happens in pico sec. ⇒ Vo – increases

Date:

TRANSFER CHARACTERISTICS OF BISTABLE MVB

1)



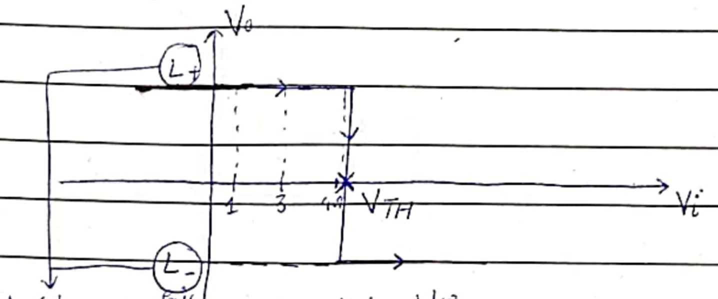
[Triggers signal not I/p]

$L_+ = 10V$

$\beta = 0.5$

$\beta L_+ = 5V$

$$\boxed{V_+ = \frac{R_1}{R_1 + R_2} \times V_o}$$



(Fig # 1)

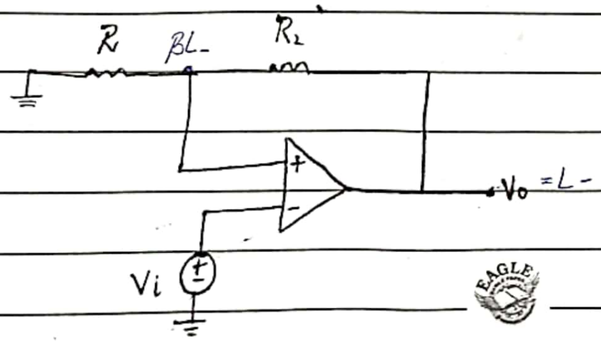
L_+ & L_- because there is no resistor or anything b/w V_o of OP-AMP & actual o/p.

Suppose,

$$V_o = L_+ \quad ; \quad V_+ = \left(\frac{R_1}{R_1 + R_2}\right) L_+ = \beta L_+$$

- $V_+ = \beta L_+$
- $V_- = V_i$
- AMP; $V_o = A(\overset{\beta L_+}{L_+} - \overset{0V}{L_-})$ $(V_i \geq \beta L_+)$
- initially, $V_- = 0V$
- We start increasing V_i gradually

2)

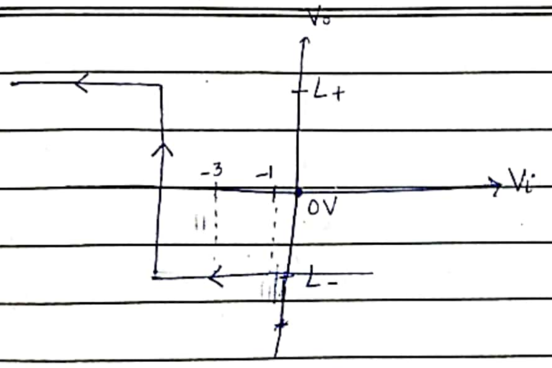


$L_- = 10V$

$\beta = 0.5$

$\beta L_- = -5V$

Date:

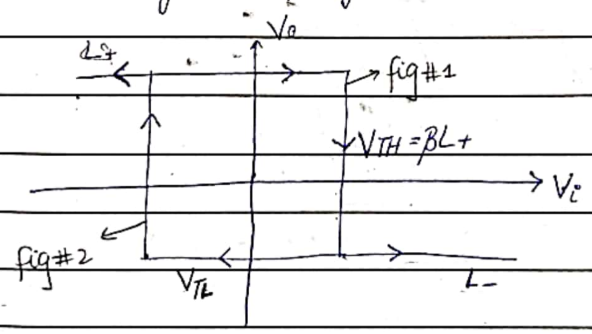


(Fig #2)

Suppose, $V_0 = L_-$; $v_+ = \frac{R_1}{R_1+R_2} \times L_- = \beta L_-$

- $V_+ = +\beta L_-$
- $V_- = V_i = 0$
- AMP ; $V_0 = A(\overset{\beta L_-}{V_+} - V_-)$ $(V_i \leq \beta L_-)$
- initially, $V_- = 0V$
- we start decrease V_i gradually

Combine Fig #1 & fig #2



~~(Hysteresis comparator)~~

V_{TH} – V Threshold high (high to low)

V_{TL} – V threshold low (low to high)

* inverting bistable MVB
or
hysterysis comparator

* $V_{TH} = \frac{R_1}{R_1+R_2} \times L_+$

$V_{TL} = \frac{R_1}{R_1+R_2} \times L_-$

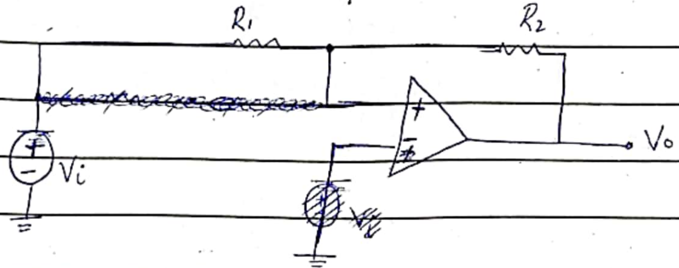
Date:

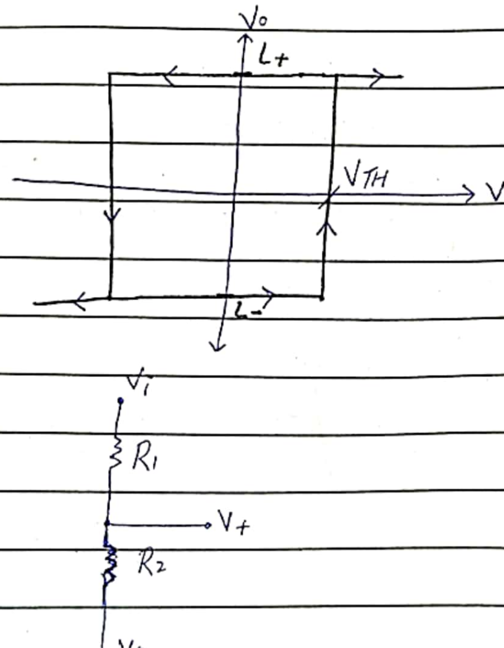
* [if trigger $(V_i) > \beta L_+$ — V_o goes from L_+ to L_-
[if trigger $(V_i) < \beta L_-$ — V_o goes from L_- to L_+

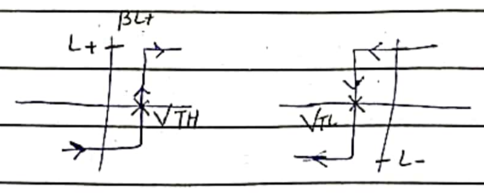
~~As you have noise at~~ ~~ripple near zero crossing~~
Hystrysis comparator:
helps to eliminate the ripples near the zero crossing of the signal wave form.

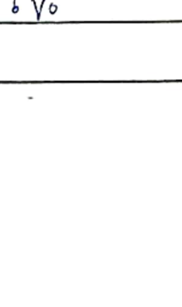
~~INVERTING BISTABLE MVB~~

24/10 NON-INVERTING BISTABLE MVB







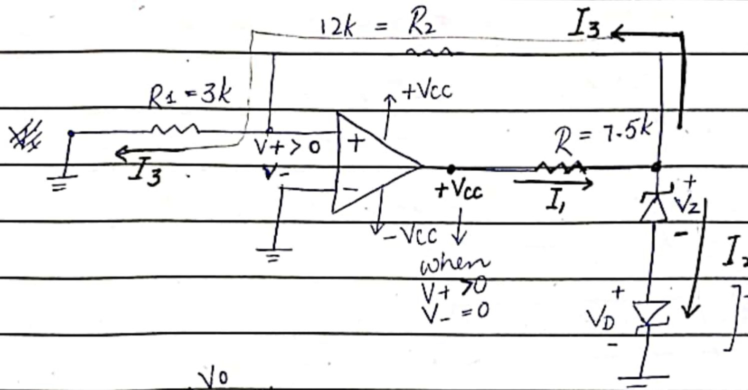


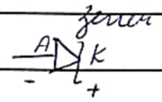
$$\star\quad V_+ = \underbrace{\left(\frac{R_1}{R_1+R_2}\right)\times V_o}_{V_i \to \text{ground}} + \underbrace{\left(\frac{R_2}{R_1+R_2}\right)\times V_i}_{V_o \to \text{ground}}$$

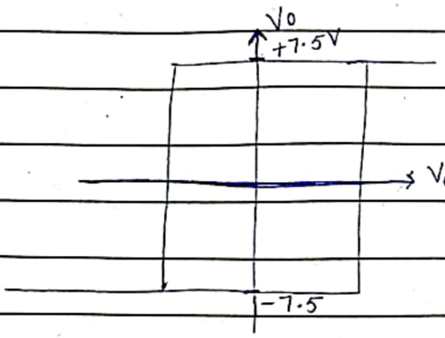
$$V_{TH} = -(R_1/R_2)\times L_-$$
$$V_{TL} = -(R_1/R_2)\times L_+$$

Date:

Q. (Non-inverting)







→ forward current in zener: it acts like an ordinary diode

$V_{AK} = 0$ (anode, cathode)

→ reverse current: it acts like a zener diode

$V = V_z$ (zener)

Let $V_z = 6.8V$, $V_D = 0.7V$ (given)

$V_o = V_z + V_D$

$= 6.8 + 0.7$

$V_o = 7.5V$ $(L+)$

for $V_+ > 0$, $V_- = 0$

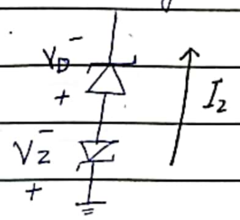
$0 - V_o = V_z + V_D$

$-V_o = 6.8 + 0.7$

$V_o = -7.5V$ $(L-)$

for $V_+ < 0$, $V_+ = 0$

(V_o V_{cc} replaced by $-V_{cc}$, direction of currents changes)



Let, $R = 1k\Omega$, $+V_{cc} = 15V$

$R_1 = 12k$, $R_2 = 3k$

$I_1 = \frac{V_{cc} - 7.5}{7.5k} = 10mA$

$I_3 = \frac{7.5 - 0}{12k + 3k} = 0.5mA$

(R_2, R_1)

Date:

$I_2 = I_3 - I_1$ $\quad \because I_3 = I_1 + I_2$

$= 0.5\,mA$

$V_+ = \dfrac{R_1}{R_1 + R_2} \times V_0$

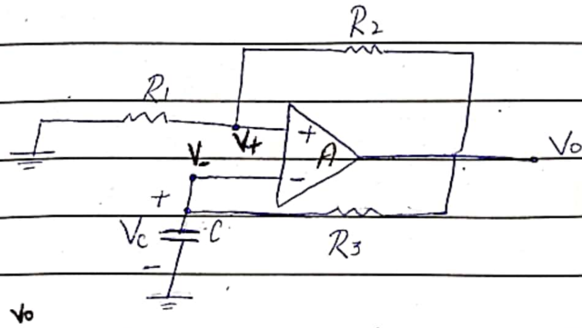
$= \dfrac{3k}{3k + 15k} \times -7.5$

$V_+ = -1.25V$

29/10

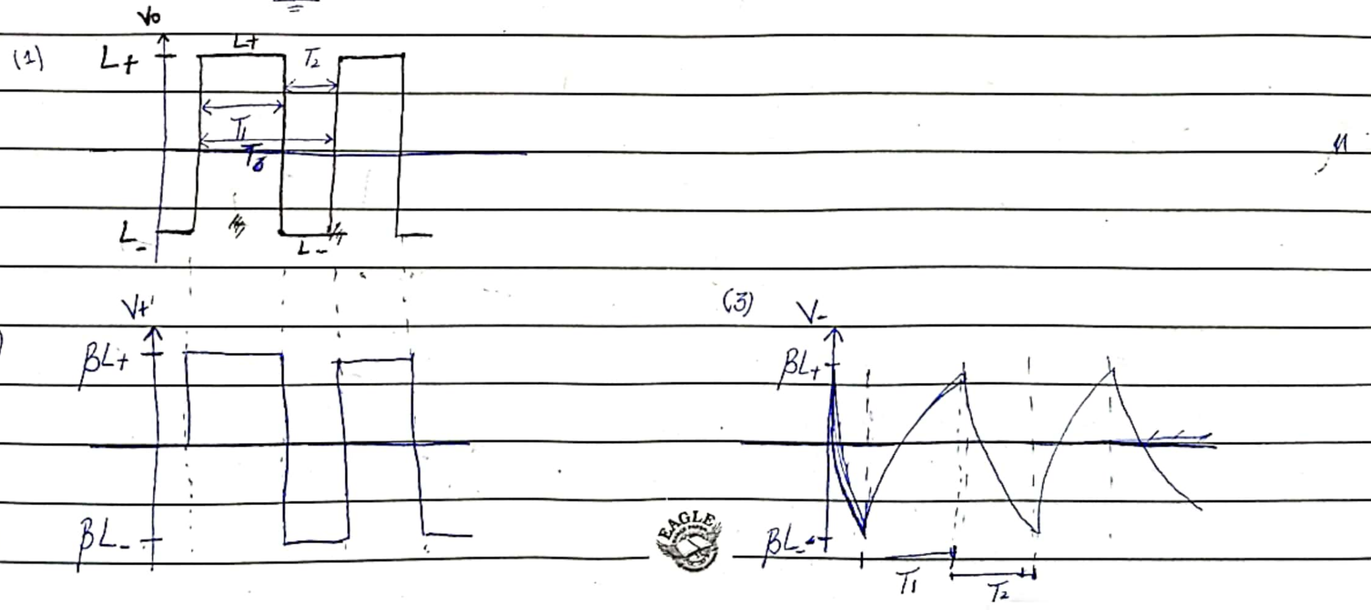
ASTABLE MVB (using inverting bistable config) (sq. or rect. wave generation)

- No stable state
- Keeps toggling between L_+ & L_-
- Used to generate sq. waves & rect. waves.
- Uses bistable MVB circuit with RC circuit in ~~+ve~~ -ve feedback.



$V_0 = A(V_+ - V_-)$

$V_- = V_C$



Date:

- Initially:

$$V_0 = L_+ \quad , \quad V_+ = \beta L_+$$

- V_c will start rising exponentially.
- V_- will start rising exponentially
- Because $V_0 = L_+$ is fixed, V_+ is also fixed
- $V_0 = A(V_+ - V_-)$

TIME PERIOD:

$$v_c(t) = v_c(\infty) - (v_c(\infty) - v_c(0))\, e^{-t/RC}$$

for T_1:

$$v_c(0) = \beta L_-$$

$$v_c(\infty) = L_+$$

$$* \because \beta = \frac{R_1}{R_1 + R_2}$$

at $t = T_1$, $v_c(t) = \beta L_+$

$$\beta L_+ = L_+ - (L_+ - \beta L_-)e^{-T_1/RC}$$

$$\ln\left(\frac{\beta L_+ - L_+}{-(L_+ - \beta L_-)}\right) = \ln\left(\frac{-T_1}{RC}\right)$$

$$* \quad T_1 = RC \ln\left(\frac{1 - \beta(L_-/L_+)}{1 - \beta}\right)$$

For symmetry power supplies ; $L_+ = -L_-$

$$* \quad T_1 = RC \ln\left(\frac{1+\beta}{1-\beta}\right)$$

for T_2:

$$v_c(0) = \beta L_+$$

$$v_c(\infty) = L_-$$

at $t = T_2$, $v_c(t) = \beta L_-$

EAGLE

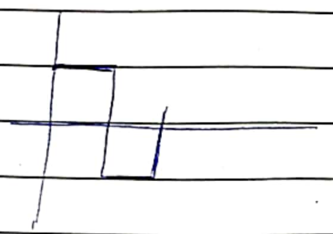
Date: ______

$\beta L_- = L_- - (L_- - \beta L_+)e^{-T_2/RC}$

* $T_2 = RC \ln\left(\frac{1-\beta(L_+/L_-)}{1-\beta}\right)$

* $T_2 = RC \ln\left(\frac{1+\beta}{1-\beta}\right)$

* for symmetrical power supplies $T_1 = T_2$

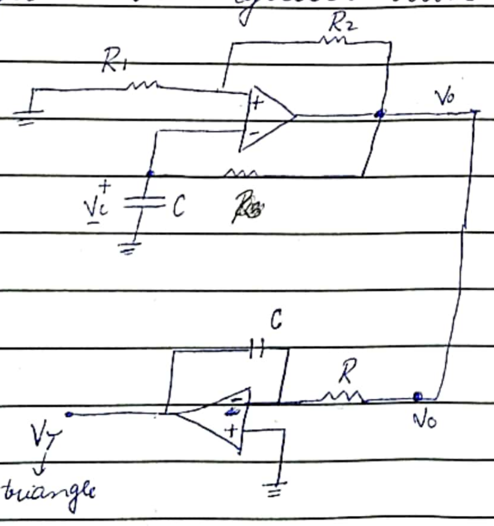


(Sq. wave)

* $\beta = \frac{R_1 \cancel{+R_2}}{R_1+R_2}$

* $T = T_1 + T_2$

To get a triangular wave :

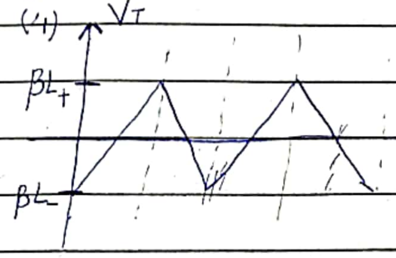


$V_T = \frac{-1}{RC}\int V_o\,dt$

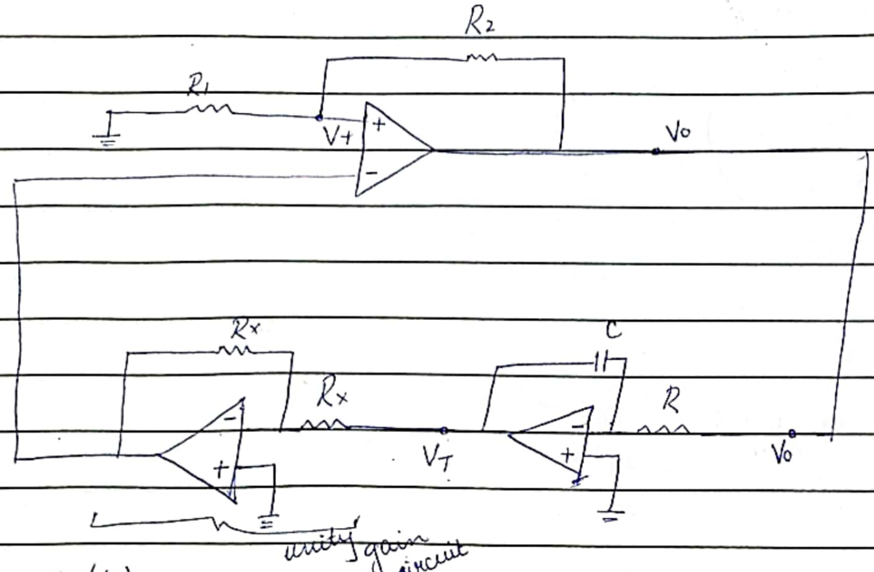
assume, $V_o = 10$

$= \frac{-1}{RC}\int +10\,dt$

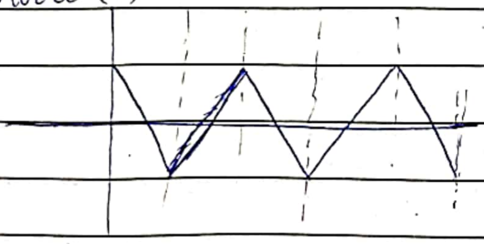
$RC = 1$



Date:



invert (4)



Function generator (Sq. + Δ wave ...)

For symm. power supplies ; $L_+ = -L_-$

$$T_1 = RC \ln\left(\frac{V_{TH} - V_{TL}}{L_+}\right)$$

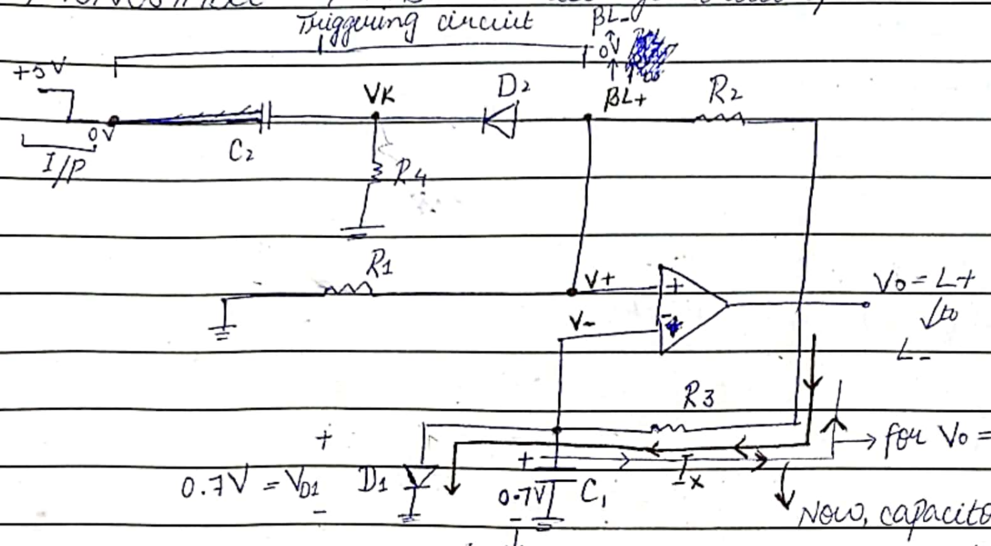
$$T_2 = RC \ln\left(\frac{V_{TH} - V_{TL}}{-L_-}\right)$$

$\because$ $V_{TH} = \beta L_+$

$\because$ $V_{TL} = \beta L_-$

Date: 31/10

MONOSTABLE MVB (Pulse generator)

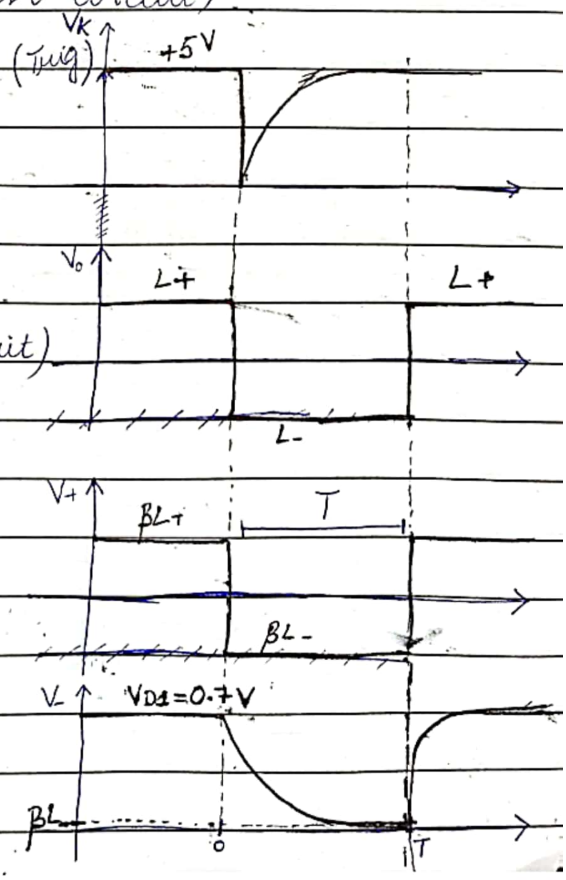


* low pass filter acts like an integrat
* high pass filter acts like a differentiator

(capacitor is charged at 0.7V) — at L+)

$\beta = 0.1$
$L_+ = 12V$
$\beta L_+ = 1.2V$

- $V_o = L_+$, $V_+ = \beta L_+$
- Because $V_K > \beta L_+$ ($\because V_K$ $5V$)
  $\Rightarrow D_2$ is reverse biased (open circuit)
- For I_x flow, D_1 is forward biased
  $\Rightarrow D_1$ is short circuit
- $V_o = A(V_+ - V_-)$
  $= A(\overset{1.2}{\beta L_+} - 0.7)$
- When apply a trigger ; $V_K = 0V$
  $\Rightarrow D_2$ becomes forward biased (~~open~~ short circuit)
- D_2 is short circuit , 0V at V_K will travel ahead & force itself at V_+
- $V_o = A(V_+ - V_-)$
  $= A(\cancel{0}\beta L_- - 0.7)$
- In this duration, V_K becomes +ve again
  $\Rightarrow D_2$ is again open



Date: ____________

$$V_0 = A(V_+ - V_-)$$
$$= A(\beta L_- - \underset{=\beta L_-}{V_-})$$
$$-1.2 - (-1.2001)$$

$\cancel{y(t)} = y\,\cancel{(\infty)}$

TIME PERIOD:

$$v_c(t) = v_c(\infty) - (v_c(\infty) - v_c(0))\, e^{-t/R_3C_1}$$

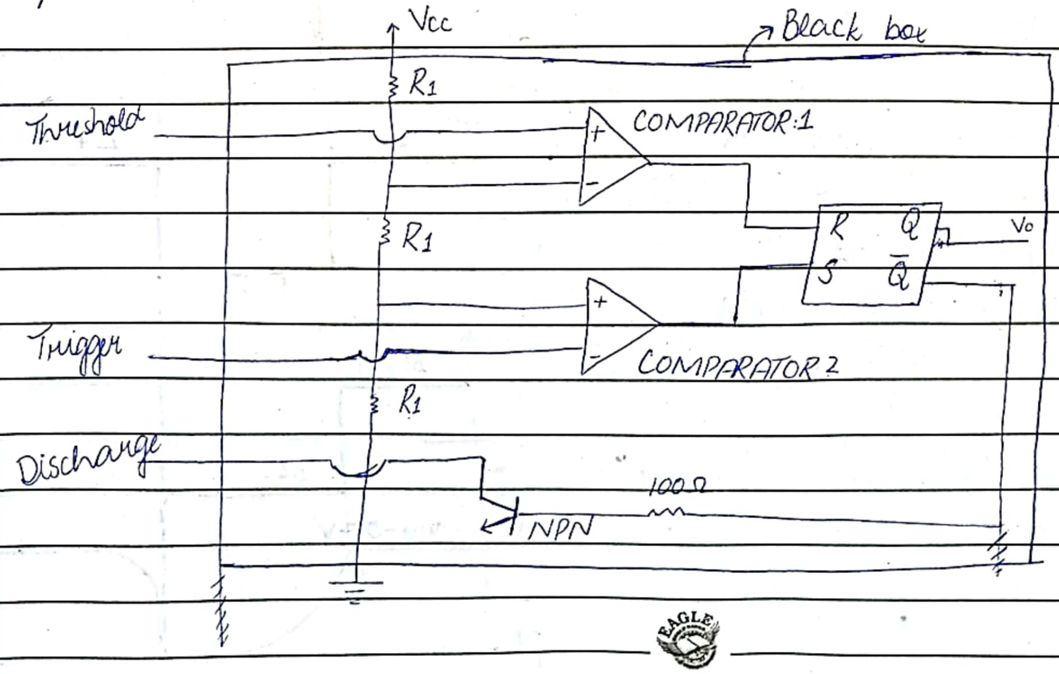
$v_c(0) = V_{D1}$ at $t = T$; $v_c(T) = \beta L_-$

$v_c(\infty) = L_-$

$$\beta L_- = L_- - (L_- - V_{D1})\, e^{-T/R_3C_1}$$

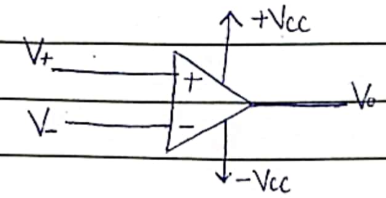
$$\Rightarrow \; * \; \boxed{T = R_3C_1 \ln\left(\frac{V_{D1} - L_-}{\beta L_- - L_-}\right)}$$

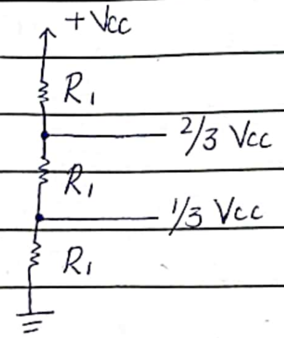
7/11 555 TIMER CIRCUIT:



Date:

COMPARATOR:

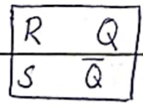




$V_+ > V_- \Rightarrow V_o = +V_{cc}$

$V_+ < V_- \Rightarrow V_o = GND$

R	S	Q	$\overline{Q}$	
0	0	Previous state		
0	1	1	0	(set)
1	0	0	1	(reset)
1	1	undefined		

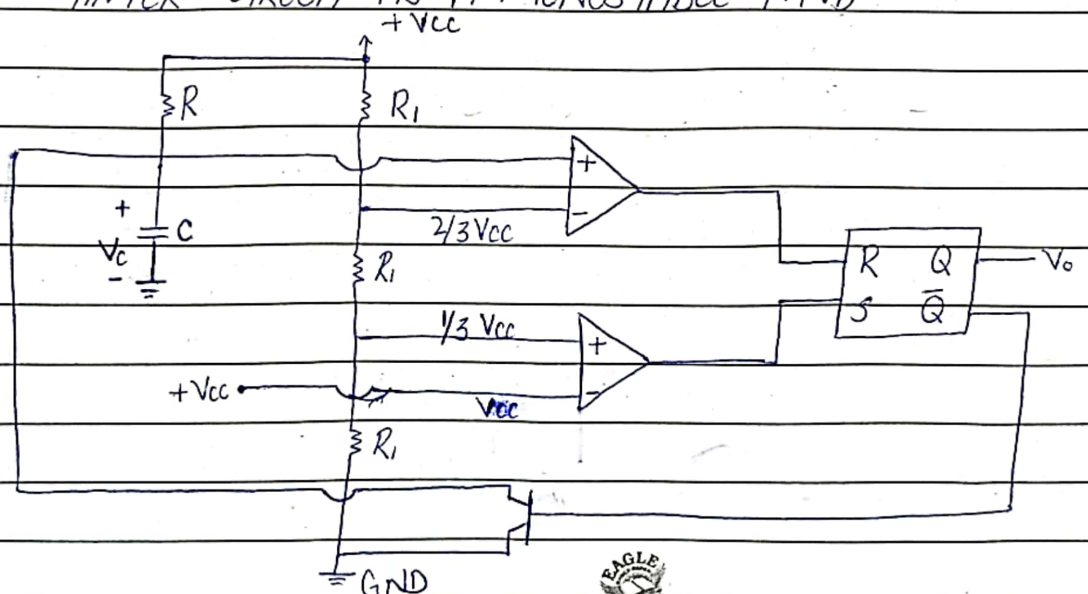


R S filip flop

when $\overline{Q} = 0$ NPN open (off)

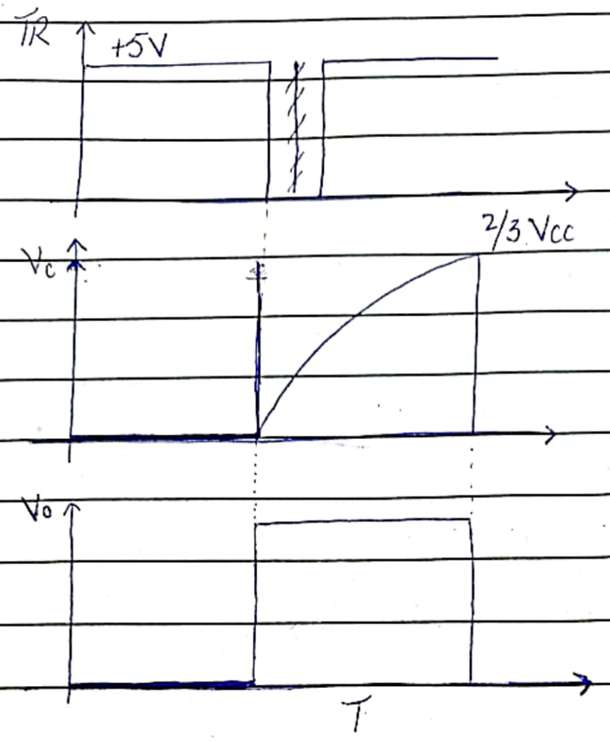
when $\overline{Q} = 1$ NPN short (ON)

555 TIMER CIRCUIT AS A MONOSTABLE MVB



Date:

Trigger (TR)

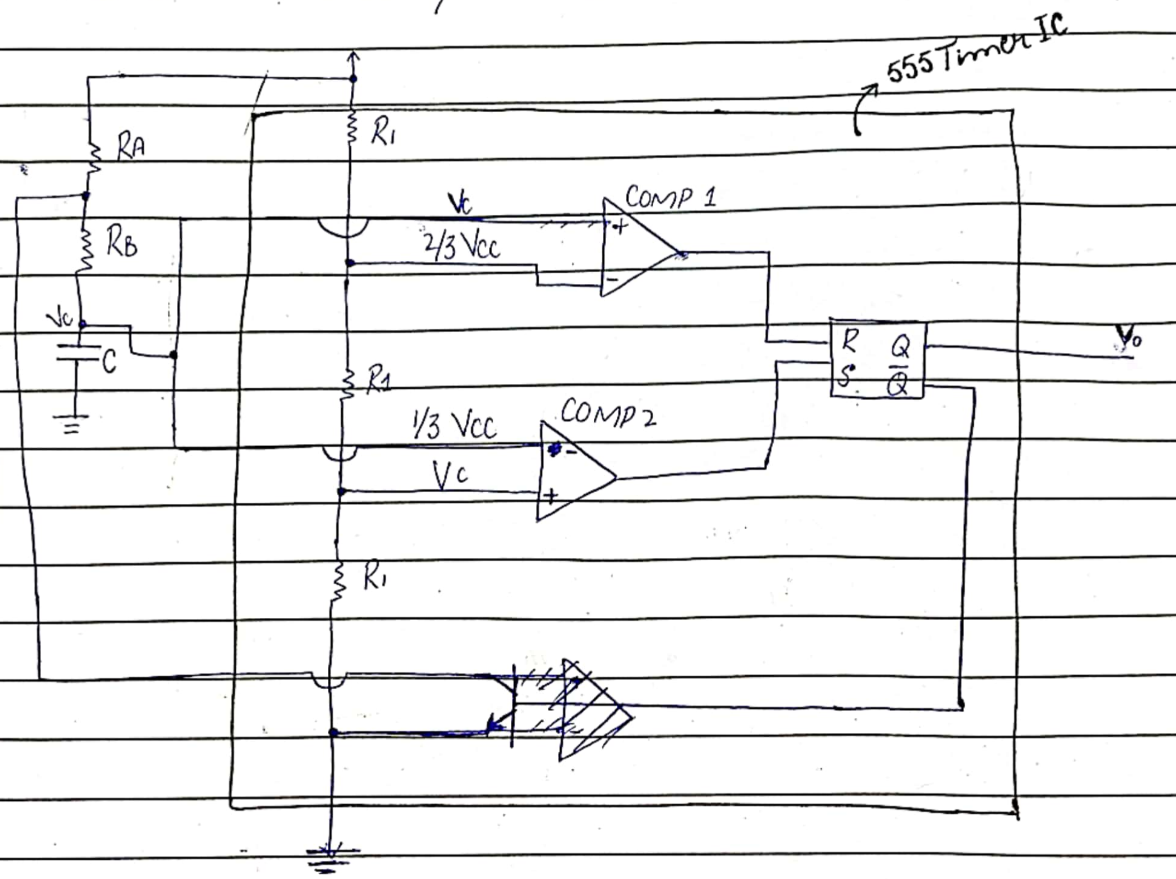


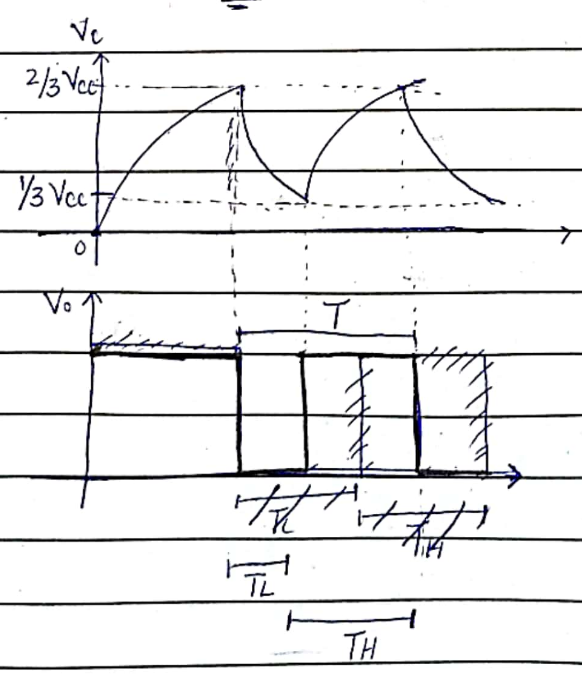
★ $T = \ln 3 RC$

$T = 1.1\, RC$

Date: 12/11

ASTABLE MVB USING 555 TIMER





flip flop

R	S	Q
0	0	Prev
0	1	1
1	0	0
1	1	∞

$\tau_c = (R_A + R_B)C$

charging $\tau_d = R_B \times C$

discharging

Date:

$v_c(t) = v_c(\infty) - (v(\infty) - v(0))\, e^{-t/RC}$

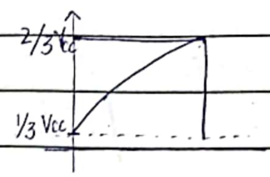
FOR CHARGING

$v(\infty) = V_{cc}$

$v(0) = \cancel{\ }\ 1/3\ V_{cc}$ (initial value)

at $t = T_H$

$\Rightarrow v_c = 2/3\ V_{cc}$



$\frac{2}{3} V_{cc} = V_{cc} - (V_{cc} - 1/3\ V_{cc})\, e^{-T_H/(R_A+R_B)C}$

* $\boxed{T_H = \ln 2\,(R_A + R_B)\,C}$

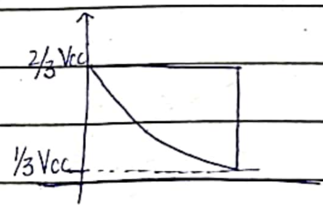
FOR DISCHARGING

$v(\infty) = 0$

$v(0) = 2/3\, V_{cc}$

at $t = T_L$

$\Rightarrow v_c = 1/3\ V_{cc}$



$\cancel{-1/3 V_{cc} = V_{cc}^{0} - (V_{cc}^{0}\ \ /}$

$1/3\, V_{cc} = 0 - (0 - 2/3\, V_{cc})\, e^{-T_L/R_B C}$

* $\boxed{T_L = \ln 2\,(R_B)\,C}$

$T = T_H + T_L$

* $\boxed{T = \ln 2\,(R_A + 2R_B)\,C}$

* $D = T_H/(T_H + T_L) = T_H/T$

* $\boxed{D = \frac{R_A + R_B}{R_A + 2R_B}}$ (duty cycle)

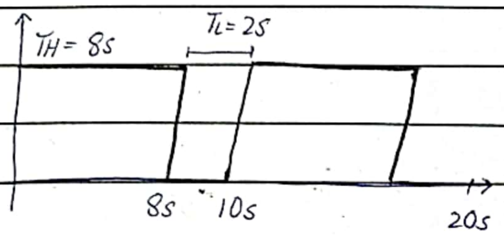
$D \geq 51\%$ (max)

Date:

DUTY CYCLE:

The percentage OR ratio of the total time period for which the signal is logic high OR logic '1'.

EXAMPLE:



$D = 8/10 = 0.8$ (80%)

* $D = \frac{T_H}{T_H + T_L} \times 100\%$

DRAWBACKS

- $T_H \neq T_L$ (never)

In current settings, T_H can never be equal to T_L. So, you can't achieve a square wave and a 50% duty cycle.

FOR 50% DUTY CYCLE:

$T_H = \ln 2\,(R_A)\,C$

$T_L = \ln 2\,(R_B)\,C$

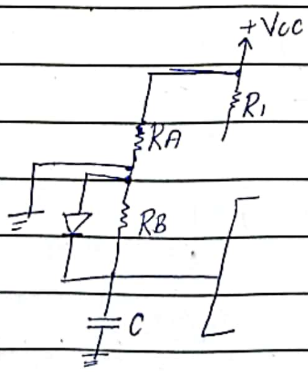
$T = T_H + T_L$

$= \ln 2\,(R_A + R_B)\,C$

$D = \frac{T_H}{T_H + T_L} = \frac{T_H}{T}$

$= \frac{R_A}{R_A + R_B}$

$\therefore R_A = R_B$



Date: 14/11

FOR DUTY CYCLE > 50%

$T_1 = \ln 2\,(R_A + R_B)C$

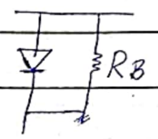
$T_2 = \ln 2\,(R_B)C$

$T = \ln 2\,(R_A + 2R_B)C$

$f = 1/T$

$D = \dfrac{R_A + R_B}{R_A + 2R_B}$

FOR DUTY CYCLE ≤ 50%



$T_1 = \ln 2\,(R_A)C$

$T_2 = \ln 2\,(R_B)C$

$T = \ln 2\,(R_A + R_B)C$

$f = 1/T$

$D = \dfrac{R_A}{R_A + R_B}$

Q. Use 1nF capacitor in a 555 timer circuit, find the value of R that generates a pulse of 10μs.

$T = (\ln 3)\,RC$

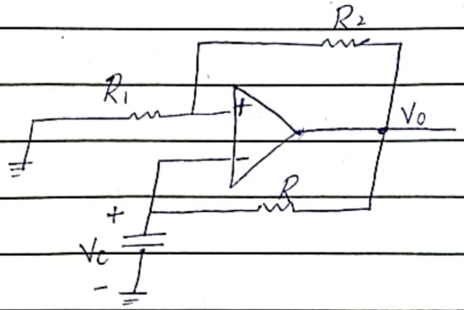
$\dfrac{10\mu}{(\ln 3)(10n)} = R$

$R = 9.1\,k\Omega$

Q. Design 2 different ~~a wave~~ circuits that can generate sq. wave. (50% duty cycle)

Date:

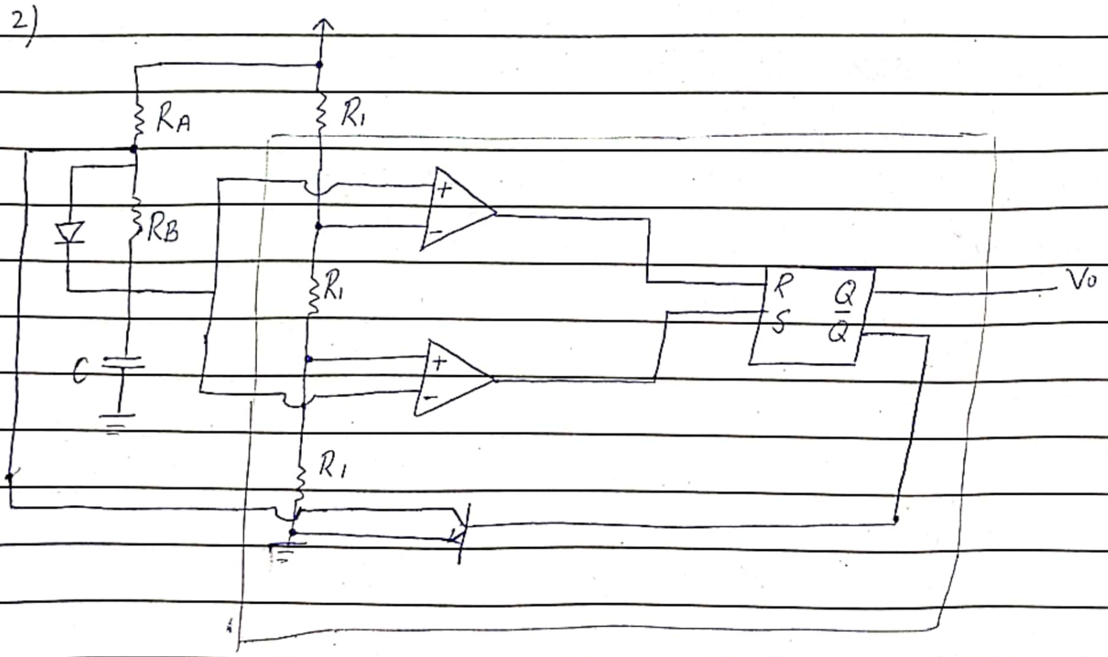
1)



$$T = RC \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$L_+ = -L_-$

2)



Q. Use a 680pF capacitor, design a circuit using 555 timer that generates a rectangular signal of 75% duty cycle & 50kHz frequency find R_A & R_B.

→ Astable MVB using 555 timer

$\cancel{f}$ $T = 1/f = 1/50k = \cancel{\quad}$ $20\mu s$

Date:

$D = \frac{R_A + R_B}{R_A + 2R_B}$ $\because 75\% = 0.75 = 3/4$

$\frac{3}{4} = \frac{R_A + R_B}{R_A + 2R_B}$

$4R_A + 4R_B = 3R_A + 3R_B$

$R_A = 2R_B$ (i)

$T = \cancel{R}\ \ln 2 \,(R_A + 2R_B)C$

$20\mu = \ln 2\,(R_A + 2R_B)(680\cancel{p})$

$R_A + 2R_B = \frac{20\mu}{(\ln 2)(680p)}$

$R_A + 2R_B = 42.432k$

$\rightarrow$ (i) $R_A + R_A = 42.432k$

$2R_A = 42.432k$

$\boxed{R_A = 21.216k\Omega}$

$\rightarrow$ (i) $\frac{R_A}{2} = R_B \Rightarrow \frac{21.216k}{2} = R_B$

$\boxed{R_B = 10.5k\Omega}$

Q. Use a 555 timer to generate a <u>sq. wave signal</u> (50%) of 10k Hz. using a capacitor of $1\mu F$.

$f = 1/T$

$T = 1/f = 1/10k = \cancel{0.1m}\ 100\mu s$

Date: ______

$$0.5 = \frac{R_A}{R_A + R_B}$$

$$0.5R_A + 0.5R_B = R_A$$

$$0.5R_A = 0.5R_B \quad (i) \quad \Rightarrow R_B = \frac{+0.5R_A}{0.5}$$

$$T = \ln 2\,(R_A + R_B)C$$

$$\frac{100\mu s}{(\ln 2)(1\mu)} = R_A + R_B$$

$$R_A + R_B = 144.26\,\Omega \quad (ii)$$

$$R_A = 144.26 - R_B$$

$$R_A = 144.26 - (+R_A)$$

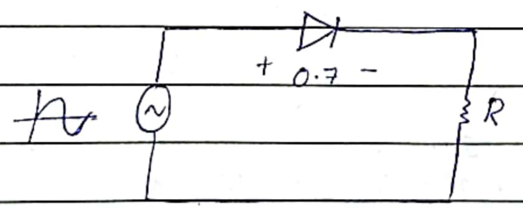
$$R_A = 144.26 - R_A \quad \Rightarrow 2R_A = 144.26\,\Omega$$

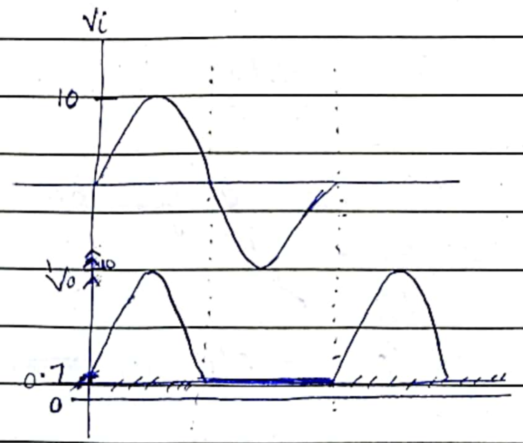
$$\boxed{R_A = 72.13\,\Omega}$$

$$R_B = 144.26 - 72.13$$

$$\boxed{R_B = 72.13\,\Omega}$$

ORDINARY HALF-WAVE RECTIFIER:



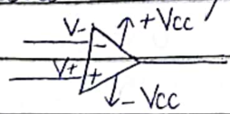


PRECISION HALF-WAVE RECTIFIER:

Date: ______

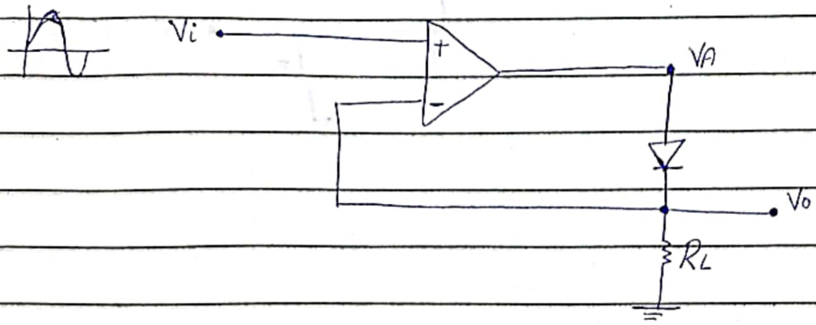
SUPER- DIODE:

Diode as comparator



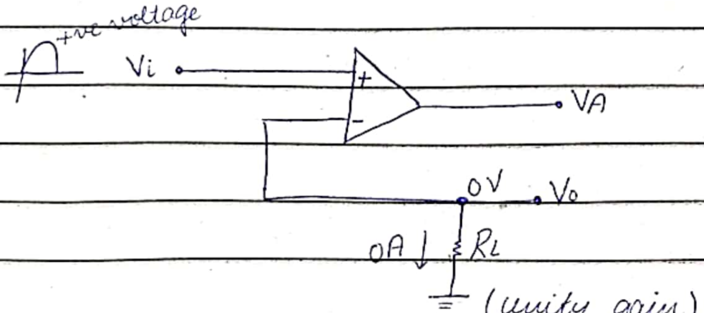
$V_+ > V_- \Rightarrow V_o = L_+$

$V_+ < V_- \Rightarrow V_o = L_-$

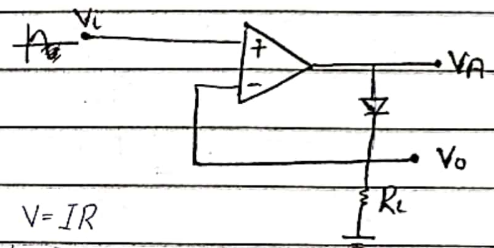


→ During the +ve half cycle ($V_i > 0$)

- For a moment, ignore that the diode is existing.



(unity gain)



$V = IR$

$V = 0$

forward biased → short circuit

$\Rightarrow^{*} V_i = V_o$

$V_+ = +ve$

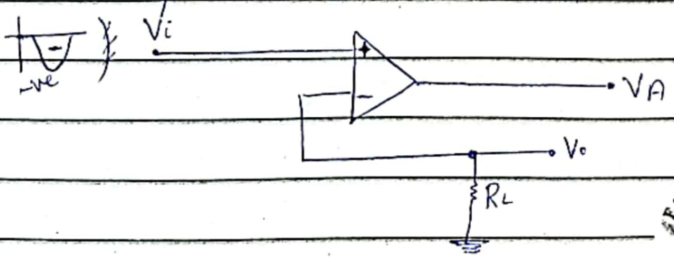
$V_- = 0$

$V_A = A(\overset{+ve}{V_+} - \overset{0}{V_-}) = +ve$

$V_A = L_+$

$\Rightarrow \boxed{V_i = V_o}$

→ DURING THE -ve HALF CYCLE ($V_i < 0$)



Date:

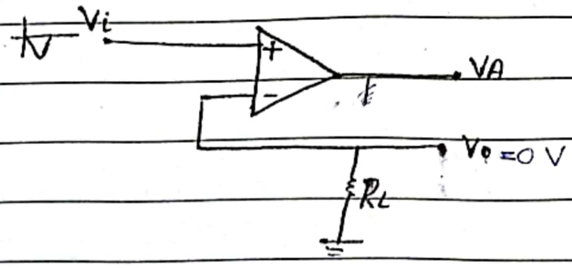
$V_A = -ve$

$V_o = 0$

$V_A = A(V_+ - V_-)$

$= A(-ve - 0) \Rightarrow -ve$

$\Rightarrow \boxed{V_o = 0V}$



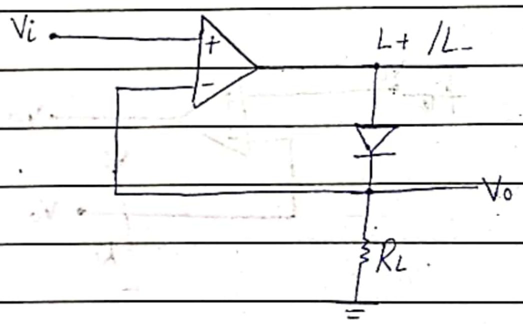
diode reverse biased

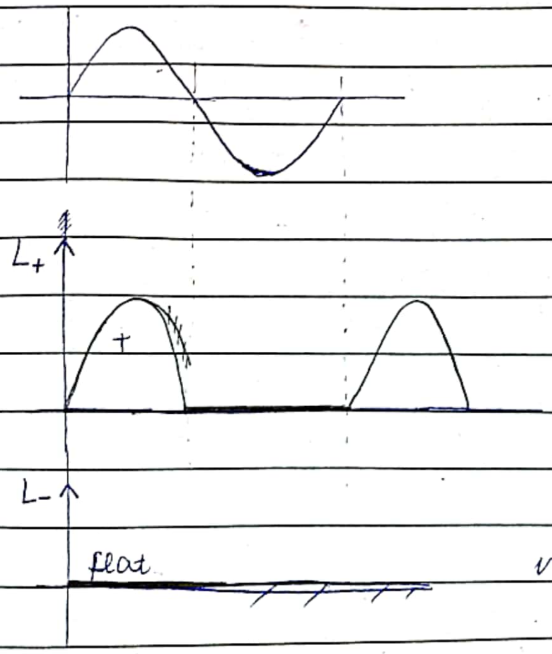
→ open circuit

19/11

ACTUAL CIRCUIT

1)



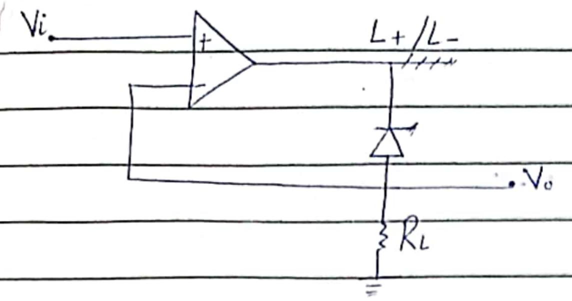


when short circuit

when open circuit

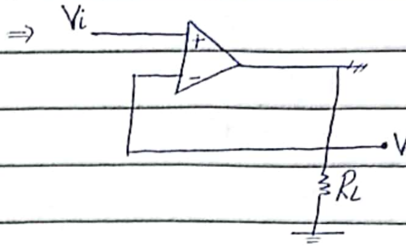
Date:

2)

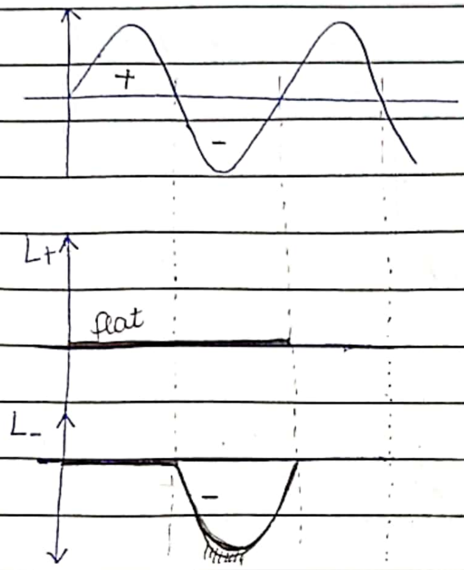


when short circuit

⇒



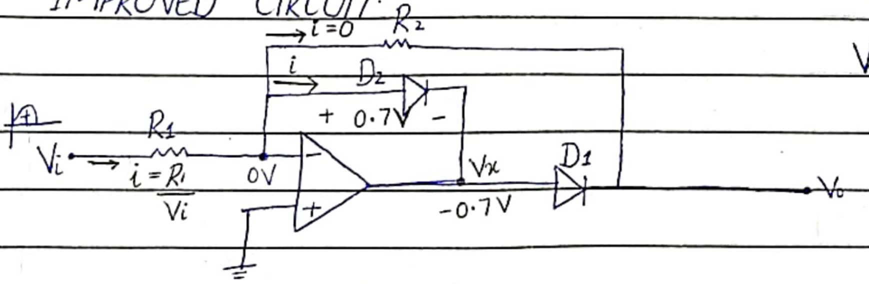
(unity gain buffer)



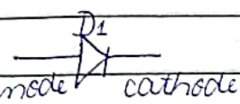
when open circuit

when short circuit (unity gain buffer)

IMPROVED CIRCUIT:



$V_i \rightarrow +ve$



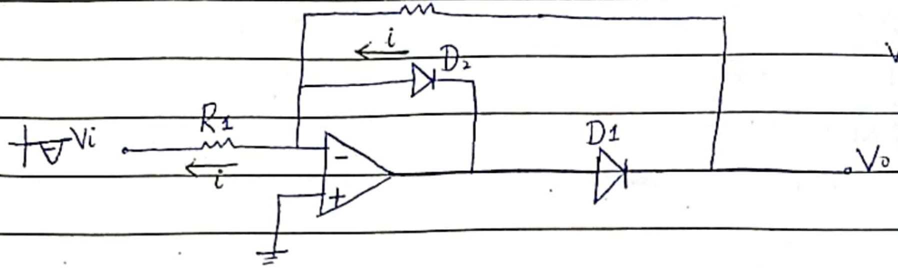
FOR POSITIVE HALF CYCLE:

- D_2 is forward biased (short circuit)
- 'i' passes through D_2.
- D_1 will be reverse biased (open circuit)

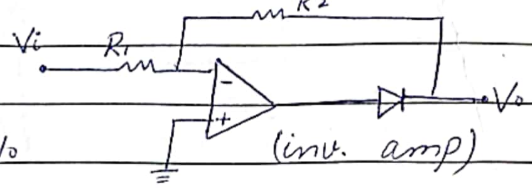
⇒ open circuit ($V_{ANODE} < V_{CATHODE}$)

Date: ______

$V_i \rightarrow -ve$



after D_2 is open

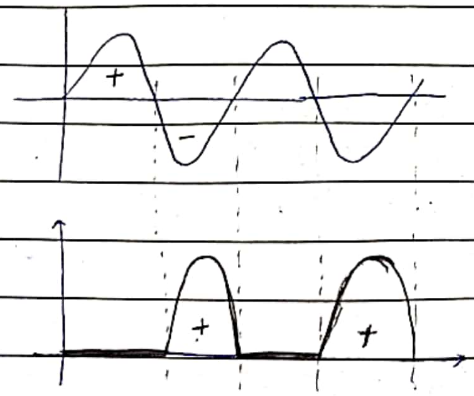


(inv. amp)

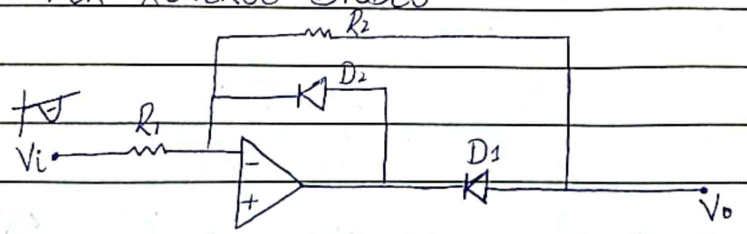
$$\Rightarrow V_o = \left(\frac{-R_2}{R_1}\right) \times V_i$$

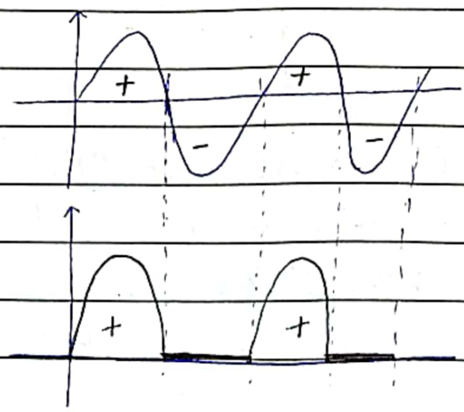
FOR NEGATIVE HALF CYCLE

- D_2 is open circuit
- 'i' passes through D_1
- D_1 is forward biased (open circuit)



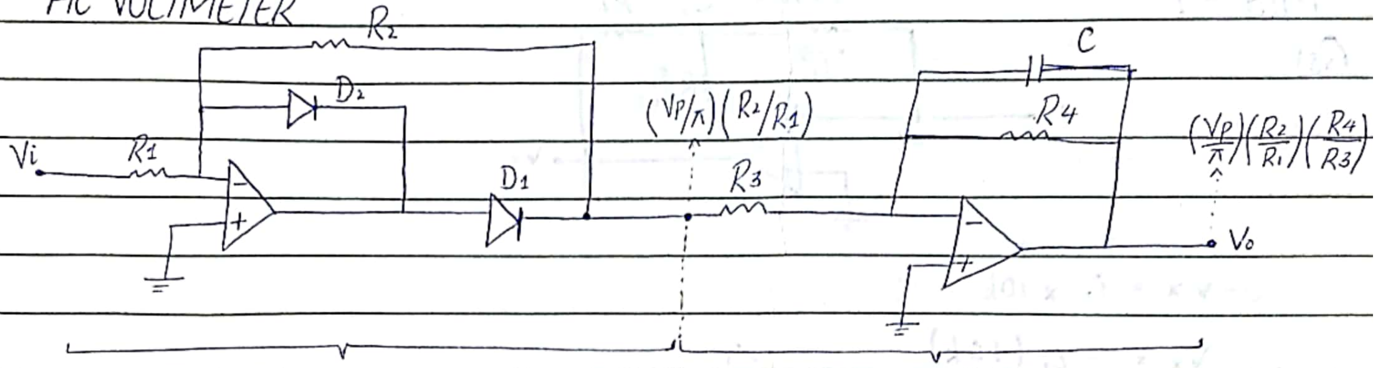
FOR REVERSE DIODES





Date: ____________

AC VOLTMETER



Precision rectifier

DC gain = R_2/R_1

Low pass filter

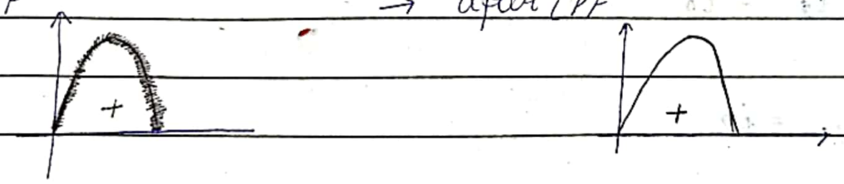
DC gain = R_4/R_3

Average value = $\frac{V_p}{\pi}$

odd harmonics → high freq. noise

LPF is used to remove noise

w/o LPF → after LPF



* $V_{o,avg} = \frac{V_p}{\pi} \times \frac{R_4}{R_3} \times \frac{R_2}{R_1}$

$\therefore V_{o,avg} \propto V_p$

* $\omega_o = \frac{1}{CR_4}$

filter design:

cut off frequency:

$\omega = 1/CR_4$

$f = 1/2\pi CR_4$

* $V_p = (\pi)(V_{o,avg})\left(\frac{R_1}{R_2}\right)\left(\frac{R_3}{R_4}\right)$

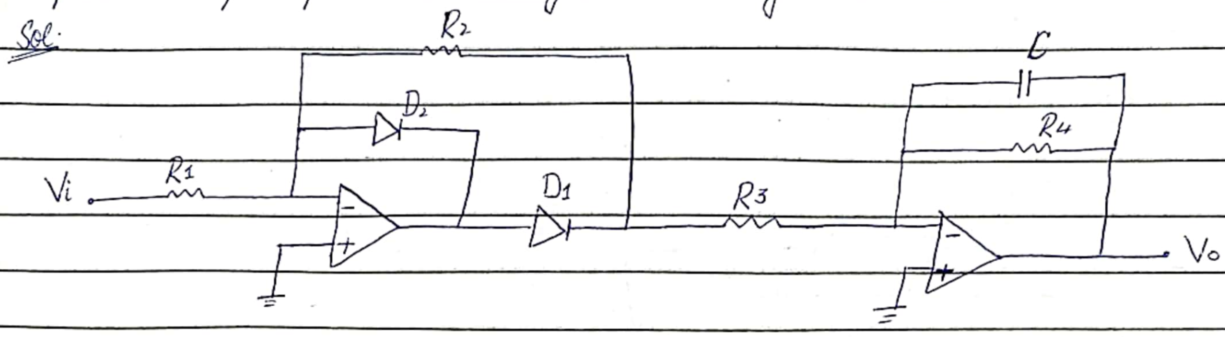
$\because V_{o,avg}$ – given

Date: 19/11

AC VOLTMETER

Q. Design an AC voltmeter for a 10Hz input signal. It is calibrated for a sine wave signal to provide a +10 V output for input of 1.0 Vrms. The Rin must as high as possible, keeping the gain of AC rectifier part to unity. The largest resistor available is 1 MΩ

Sol.



AC rectifier gain is unity $\Rightarrow \frac{R_2}{R_1} = 1.0$

input resistor must be as large $\Rightarrow R_1 = 1.0 M\Omega$

$\therefore R_2 = 1.0 M\Omega$

$V_{in} = 1.0 V_{rms}$

$= (1.0)(\sqrt{2})$

$\because V_{rms} = V_P/\sqrt{2}$

$\Rightarrow V_P = V_{rms} \times \sqrt{2}$

$$V_o = \frac{V_P}{\pi} \times \frac{R_2}{R_1} \times \frac{R_4}{R_3}$$

$$+10 = \frac{\sqrt{2}}{\pi}(1)\left(\frac{R_4}{R_3}\right)$$

* if R_3 is 1MΩ then R_4 would become ~~less~~ larger than 1MΩ.

Highest is 1.0M ⇒ let $R_4 = 1.0 M\Omega$

So, that R_3 is lesser than R_4

$$10 = \frac{\sqrt{2}}{\pi}(1)\left(\frac{1 \times 10^6}{R_3}\right)$$

$R_3 = 45 k\Omega$

$\omega = 1/CR_4$

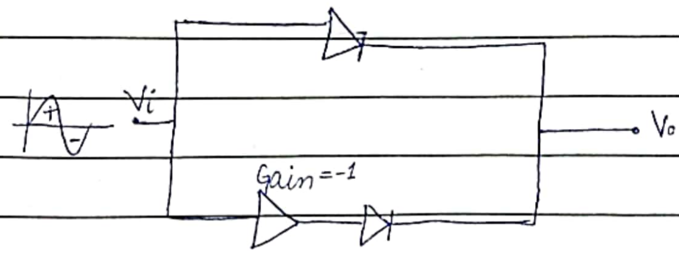
$f = 1/2\pi CR_4$

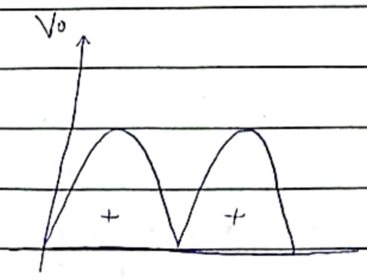
$10 = 1/2\pi C(10^6)$

$\boxed{C = 15.9 nF}$

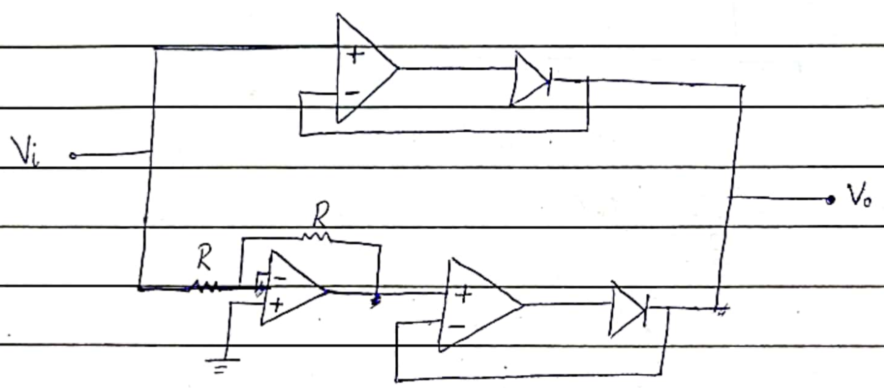
Date: 21/11

PRECISION FULL WAVE RECTIFIER:

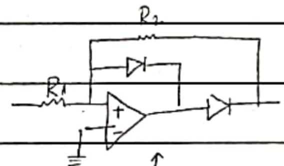




Full-wave rectifier

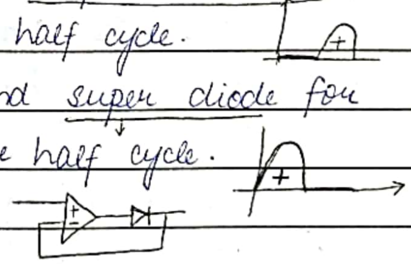


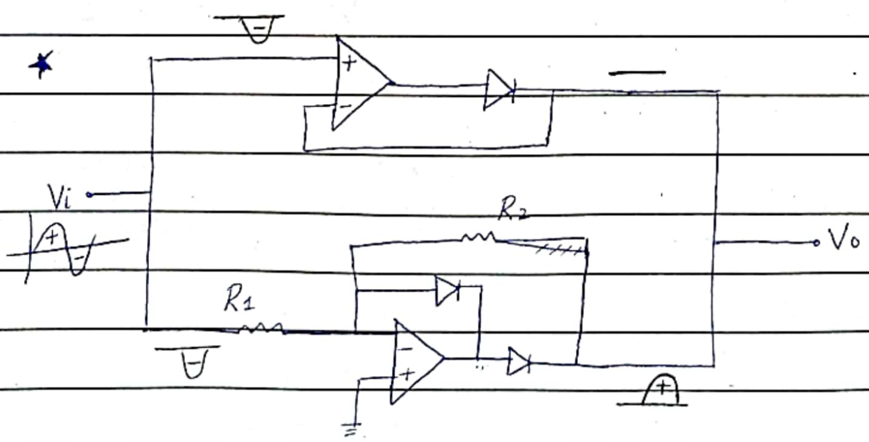
⇓



use improved circuit for -ve half cycle.

and super diode for +ve half cycle.

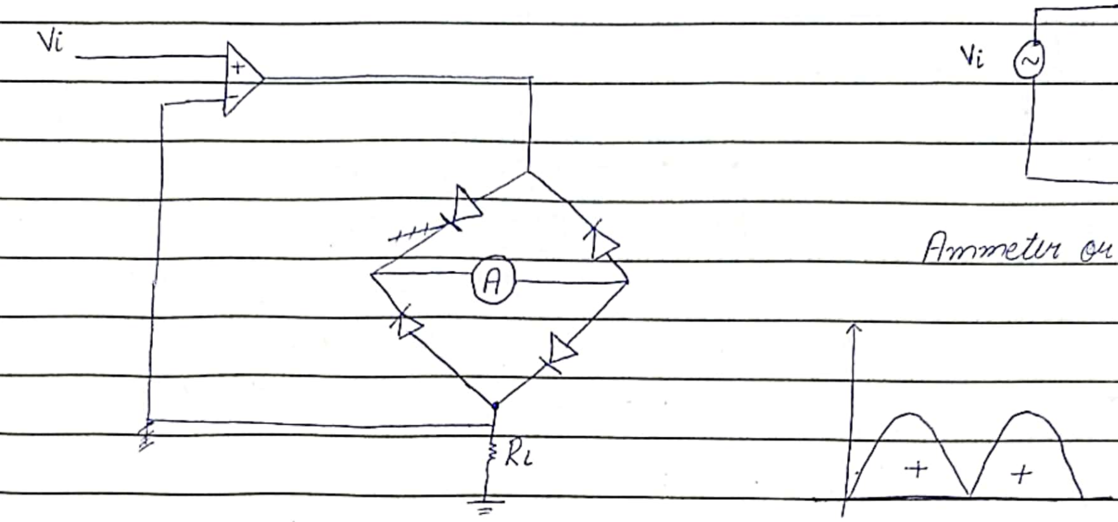




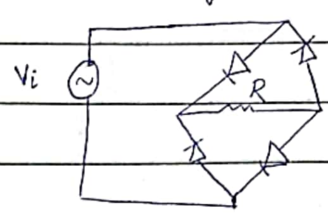
(precision full-wave rectifier)

Date:

full wave rectifier using BRIDGE (IDEAL)



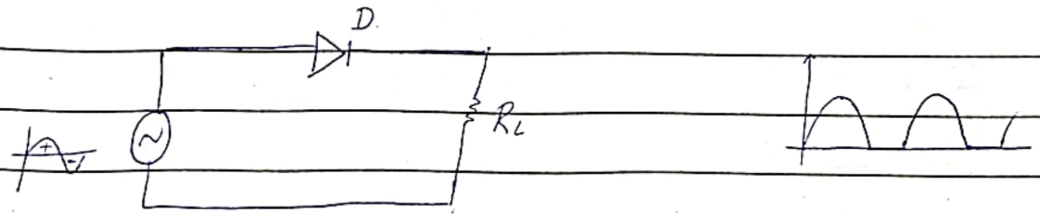
bridge

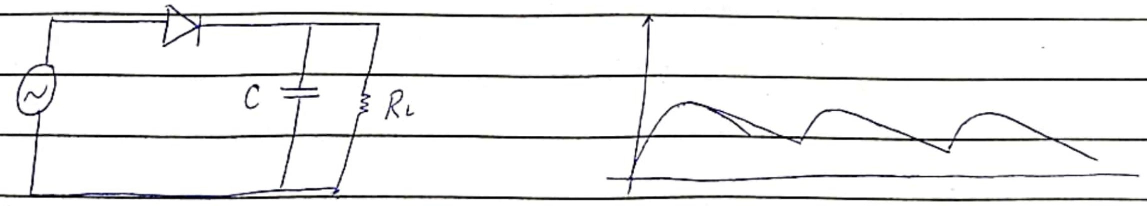


Ammeter or 0.1 Ω.

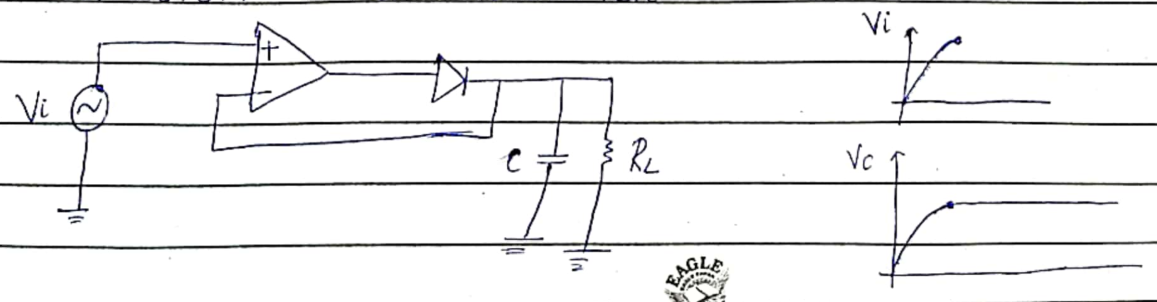
PEAK RECTIFIER

(Half wave)



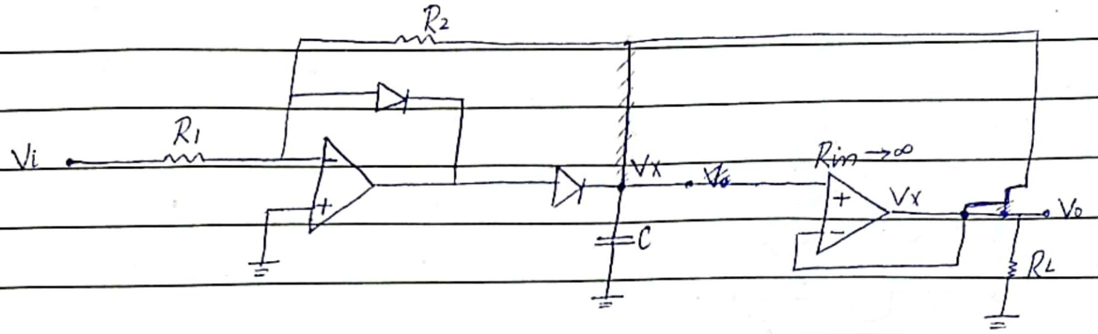


PRECISION IN PEAK RECTIFIER

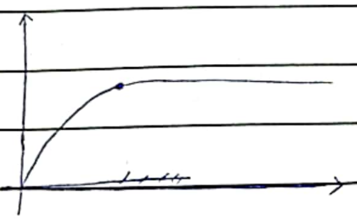


Date:

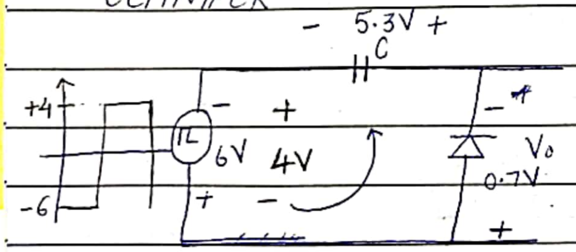
BUFFERED PRECISION RECTIFIER



* feedback loop is closed where R_L is connected.

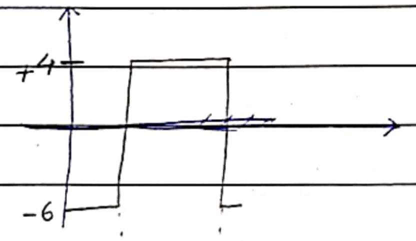


CLAMPER

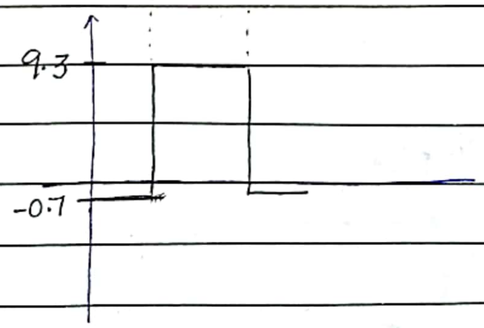


for '−6V'

for '+4V'



overall



10 − 0.7 = 9.3V

Date:

PRECISION CLAMPER